ARTEMIS II LUNAR SCIENCE AND OPERATIONS OVERVIEW. K.E. Young[1] and the Artemis II Lunar Science Team, [1]NASA Goddard Space Flight Center, 8800 Greenbelt Road, Greenbelt, MD, 20771 (kelsey.e.young@nasa.gov).

Introduction: The Artemis II mission, planned for an early 2026 launch, will bring the first crewed observations of the Moon 54 years, ushering in a new era of crewed lunar exploration. The mission includes a series of utilization objectives across human research, biological and physical sciences, lunar science, and more; this abstract focuses on the lunar science plan for the mission.

As the first crewed flight of the Orion vehicle, early mission phases include vehicle checkouts in proximity to Earth. After the trans-lunar injection burn, however, the crew will complete the transit to the Moon, where they will perform a lunar flyby around the lunar farside. During the flyby, the crew will have several hours to observe and image the Moon.

The Artemis II Lunar Science Team (LST) has developed 10 lunar science objectives [1] and four exploration capability objectives, all of which were approved by NASA's Science Mission Directorate in October 2024 (Table 1).

Lunar Science Objectives	Priority
Color Provinces & Albedo Variations	1
Impact Flashes	1
Photometric Changes	2
Landing Sites & Lunar Poles	2
Impact History	2
Dust & Exosphere	2
Tectonics	2
Volcanic History	3
Terminator & Limb	3
Earth From Space	3

Exploration Capability Objectives
Exercise Crew Lunar Imaging /Observation Product Development & Execution
Exercise Science Flight Team Support
Exercise Science Data Capture Methods
Exercise Science Data Archiving Methods

Table 1: The SMD-approved Artemis II 10 Lunar Science and 4 Exploration Capability Objectives for Artemis II lunar science. Lunar Science Objective Priority was determined based on the ability of the Artemis II mission profile to accomplish our objectives.

Artemis II Lunar Science Mission Milestones: In addition to the several hour flyby, during which Orion will be pointed to support direct crew observations of the Moon, the crew will complete several other activities pre- and post-flyby to support the lunar science mission objectives (Fig. 1). As the LST cannot generate the final list of flyby targets the crew will be asked to observe and document until after launch [1], the crew is given time prior to the lunar flyby to review the Lunar Targeting Plan [2], which displays their flyby timeline and objectives. The crew will communicate with Mission Control and the Science Officer both before and after the flyby via two planned crew conferences focused on the flyby itself.

Artemis II Lunar Science Data: Artemis II lunar science objectives depend on both crew observations and Orion vehicle camera data. The crew will collect three types of data: **(1)** crew descriptions of lunar targets, recorded as audio files via their Portable Computing Devices (PCDs); **(2)** images captured via a handheld Nikon D5 DSLR camera equipped with a 80-400mm zoom lens; and **(3)** annotations captured via OneNote files on their PCDs (analogous to a field geologist's field notebook). The crew has received extensive training to support these observations in the classroom, field environments, integrated flight control team simulations, and the Orion vehicle mockup located in the NASA Johnson Space Center (JSC) Space Vehicle Mockup Facility [3].

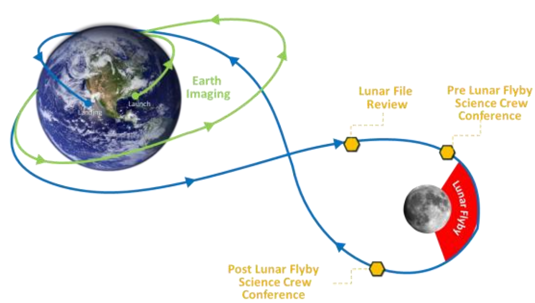

Figure 1: The Artemis II flight plan showing major lunar science activities. All timing is approximate. Opportunistic imaging is also planned for both the approach to and journey back from the Moon.

Artemis II Lunar Science Operations Support Structure: The execution of the Artemis II lunar science mission objectives will be led by the crew in the Orion spacecraft and supported by the Flight Control Team based at the Mission Control Center (MCC) at the NASA Johnson Space Center (JSC).

Artemis II will be the first mission where the Science Officer will be on console in the Flight Control Room (FCR, or main front room in MCC; Fig. 2). The Science Officer is a new flight controller position, responsible for each Artemis mission's lunar science and geology objectives. The Artemis II Science Officers are supported by the LST, who will populate two support facilities.

The Science Evaluation Room (SER) is the primary backroom for lunar science and geology, led by the SER Lead and Deputy Lead and populated by lunar scientists [1], visualization and lunar data experts, observation planners, science documentation specialists, and more. The voice up and out of the SER, to the Science Officer and others, is the SERCOM (SER Communicator). The SER, a flight control room in the main MCC building at JSC (Building 30) was designed and developed for Artemis and future Mars missions (Fig. 3). It has been operational since June 2025, and has been used in a number of Artemis II and III mission simulations.

The Science Mission Operations Room (SMOR) supports the SER by undertaking early data processing and delivering to the SER packaged data for analysis. The SMOR is operated by the Astromaterials Research and Exploration Science (ARES) group at NASA JSC (Building 36, Fig. 4) and has been operational since Spring 2025.

Figure 2: The Science Officer console in the Artemis Flight Control Room.

Figure 3: The Science Evaluation Room (JSC Building 30) in use by the LST during an Artemis II mission simulation.

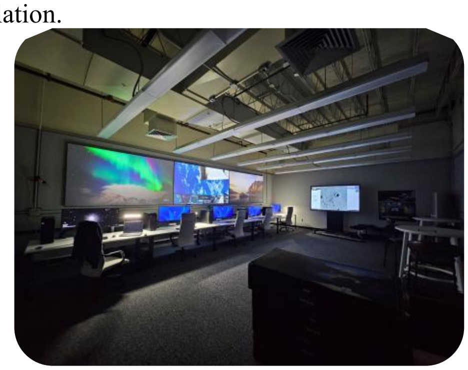

Figure 4: The Science Mission Operation Room, located in Building 36 at NASA JSC.

Post-Mission Plans: Within six months of the mission, four items will be made available to the public:

1. Post Mission Lunar Science Report summarizing preliminary science results;
2. Artemis II Lunar Science Operations Report summarizing the structure, processes, and tools used by the LST during operations;
3. Artemis II Lunar Science Data User's Guide to provide the community with the knowledge they need to access and use the data in the data archive [4];
4. All Artemis II Lunar Science Data archived via the Planetary Data System [4].

Acknowledgments: We acknowledge the contributions of the entire Artemis II Lunar Science Team, the Artemis II Flight Control Team, the Artemis II crew, and the M2M community for their support for our Artemis II lunar science objectives.

References: [1] Deutsch, A. et al. (2026) *LPSC 2026.* [2] Schmidt, S. et al. (2026) *LPSC 2026.* [3] Evans, C. A. et al. (2026) *LPSC 2026.* [4] Hollibaugh-Baker, D. et al. (2026) *LPSC 2026.*